Memo: "Customers who bought this also bought..." and customer groups

To: Sarah Okafor, Head of E-Commerce | From: Data & ML team | Re: Q4 planning

Recommendation

Trial the SVD recommender and marketing groups. Calculated factor storage is 7.3 MB, compared with 161 MB for a dense customer-by-product table. On unseen purchases, 36.6% of 1,913 customers received at least one correct product in their top 10, against 27.0% for best sellers (about +/-2 percentage points).

Measure	Reference	SVD model
Calculated storage	161 MB dense table	7.3 MB factors (22x smaller than dense)
Model stage time	Original prototype not re-timed	0.7 s: fit, product similarities, score 1,913 test customers
Customers with a held-out product in their top 10	Best sellers: 27.0%	Factors + popularity: 36.6%

These are storage calculations, not saved model-file sizes. The sparse input is already only 4.99 MB, smaller than the factors. Timing excludes loading, cleaning, matrix construction, saving, and scoring customers outside the test group; measure the full production pipeline separately.

Method and why

We compress the customer-by-product table with **SVD** (a way of finding 100 hidden "taste patterns", such as "buys cake cases and doilies"), then suggest products that sit close to what a shop already buys. We chose SVD over its close cousin PCA because it works directly on a table that is 98% empty and gives product patterns in one step. We ran both and they performed the same here (36.6% vs 36.4%); our choice is based on its natural fit to sparse purchase data. Customer scores also blend in popularity: patterns alone scored 22.5%, below best sellers. Here, best sellers means products bought by the most training customers. The blend and pattern count were chosen on earlier validation data. The product-page box uses product similarity alone; the customer hit rate above does not directly measure that box.

What we lose by compressing

The 100 patterns retain 44.6% of measured variation (relative reconstruction error 72.8%). That sounds large, but what matters is suggestion quality. Comparison: a version that keeps the full table and never compresses scores 32.1% on patterns alone versus 22.5% for ours, so compression does cost some precision; the popularity nudge more than makes up for it.

Customer groups for marketing

We projected customers onto two components, then clustered those same coordinates into 3 groups: **Mixed gifts and baking shops** (46%: Vintage Snap Cards, Retrospot Tea Set Ceramic 11 Pc...); **Heart and home-decor shops** (30%: Heart Ivory Trellis Large, Wooden Frame Antique White...); **Jumbo-bag buyers** (24%: Jumbo Bag Apples, Jumbo Bag Spaceboy Design...); UK buyers make up 85-97% of each group. The map simplifies buying patterns, so use these as campaign themes and validate them with Marketing.

What could go wrong

(1) The data ends Dec 2011 and has only ~5,500 customers with history; seasons and tastes shift, so retrain nightly and re-check the hit rate monthly. (2) New customers and new products need a best-seller fallback. (3) Hit rate counts only purchases we can see; a 'miss' may still be a good suggestion. A live A/B test is the real proof. (4) Returns and cancellations were excluded, and about a fifth of order lines had no customer ID and were dropped. (5) No customer IDs appear in this memo or the slides; groups are anonymous.

Next steps

Run an A/B test on a few product pages; add filters (in stock, same price band); add recent purchases with more weight; agree one campaign per group with Marketing.